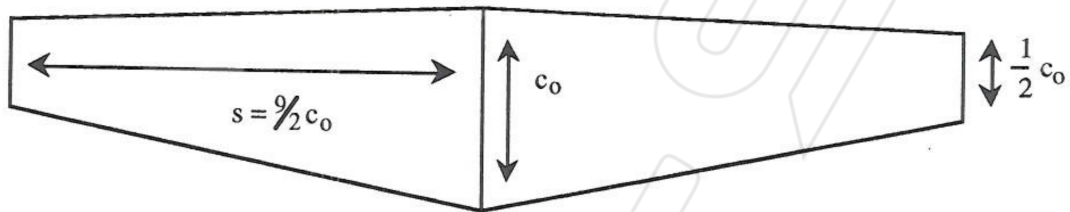


AERO60001 Aerodynamics 3: Tutorial Sheet 3

1. A plane (uncambered) tapered wing has the planform shown below. Assuming that the section lift-curve slope $a_0 = 6$ for all aerofoil sections, calculate the distribution of twist $\alpha(y)$ required to give the wing an elliptic load distribution at an overall lift coefficient of $C_L = 0.5$.



answer: $\alpha(y) = \frac{1}{4\pi} \left[\frac{1}{6} + \frac{\sqrt{1-y^2/s^2}}{1-|y/2s|} \right]$

$$C = C_0 \left(1 - \frac{1}{2} \left| \frac{y}{s} \right| \right)$$

$$\bar{C} = \int_0^s C dy = \int_0^s C_0 \left(1 - \frac{1}{2} \left| \frac{y}{s} \right| \right) d\left(\frac{y}{s}\right) = C_0 \left[\frac{y}{s} - \frac{1}{4} \left(\frac{y}{s} \right)^2 \right]_0^s = \frac{3}{4} C_0$$

$$R = \frac{2s}{\bar{C}} = 12$$

When $C_L = 0.5$, $\pi R A_1 = 0.5 \quad \therefore A_1 = \frac{1}{24\pi}$

$$\therefore \Gamma_0 = 4s U_\infty A_1 = \frac{s U_\infty}{6\pi}$$

$$\therefore \Gamma = \Gamma_0 \sqrt{1 - \left(\frac{y}{s} \right)^2} = \frac{s U_\infty}{6\pi} \sqrt{1 - \left(\frac{y}{s} \right)^2}$$

$$\therefore C_L(y) = \frac{2\Gamma(y)}{U_\infty C} = \frac{s}{3\pi C} \sqrt{1 - \left(\frac{y}{s} \right)^2}$$

$$\therefore C_L(y) = a_0 a_e(y)$$

$$\therefore a_e(y) = \frac{s}{18\pi C} \sqrt{1 - \left(\frac{y}{s} \right)^2}$$

$$W = -U_\infty \frac{C_L}{\pi R} = -U_\infty \frac{1}{24\pi}$$

$$\therefore \alpha_i = -\frac{W}{U_\infty} = \frac{1}{24\pi}$$

$$\therefore \alpha(y) = \alpha_e(y) + \alpha_i$$

$$= \frac{S}{18\pi C} \sqrt{1 - \left(\frac{y}{S}\right)^2} + \frac{1}{24\pi}$$

$$= \frac{\frac{9}{2}C_0 \sqrt{1 - \left(\frac{y}{S}\right)^2}}{18\pi C_0 \left(1 - \frac{1}{2}\left|\frac{y}{S}\right|\right)} + \frac{1}{24\pi}$$

$$= \frac{1}{4\pi} \left(\frac{\sqrt{1 - (y/S)^2}}{1 - |y/2S|} + \frac{1}{6} \right)$$

$$\bullet L = \rho U_\infty \int_{-s}^s \Gamma(y) dy$$

$$= 2\rho U_\infty s \Gamma_0 \int_0^1 \sqrt{1 - \left(\frac{y}{s}\right)^2} d\left(\frac{y}{s}\right)$$

$$\text{let } \frac{y}{s} = \sin x, \quad d\left(\frac{y}{s}\right)/dx = \cos x$$

$$\therefore L = 2\rho U_\infty s \Gamma_0 \int_0^{\frac{\pi}{2}} \frac{\cos x}{\cos x} dx$$

$$= 2\rho U_\infty s \Gamma_0 \left(\frac{\pi}{2}\right) = \rho U_\infty \pi \Gamma_0$$

$$C_L = \frac{L}{\frac{1}{2}\rho U_\infty^2 (2\bar{c} \cdot s)} = 0.5$$

$$\therefore \Gamma_0 = \frac{U_\infty \bar{c}}{\pi} = \frac{3U_\infty C_0}{4\pi} = \frac{U_\infty s}{6\pi}$$

2. A plane rectangular wing with sectional lift-curve slope $a_0 = 6$ and aspect ratio $R = 9$ is set at an incidence α to a low speed flow U_∞ . Assuming that all the coefficients above the fourth in the spanwise loading Fourier series can be neglected, obtain the lift/induced drag ratio when $\alpha = 10^\circ$.

answer: 32.4

$$C = \bar{C}$$

$$\therefore R = \frac{4s}{S} = \frac{2s}{\bar{C}}$$

$$\therefore s = \frac{R\bar{C}}{2} = \frac{9}{2}C$$

$$\therefore \mu = \frac{a_0 C}{8s} = \frac{1}{6}$$

Let $y = s \cos \theta$, and

$$\theta_1 = \frac{1}{5}\pi, \theta_2 = \frac{2}{5}\pi, \theta_3 = \frac{3}{5}\pi, \theta_4 = \frac{4}{5}\pi$$

By symmetry, $A_2 = A_4 = 0$

$$\text{LLE at } \theta_1, A_1 \sin(\theta_1) \left[\frac{1}{6} + \sin(\theta_1) \right] + A_3 \sin(3\theta_1) \left[\frac{1}{2} + \sin(\theta_1) \right] = \frac{1}{6} \left(10 \cdot \frac{\pi}{180} \right) \sin(\theta_1)$$

$$\therefore 0.4435 A_1 + 1.0345 A_3 = 0.01710$$

$$\text{LLE at } \theta_2, A_1 \sin(\theta_2) \left[\frac{1}{6} + \sin(\theta_2) \right] + A_3 \sin(3\theta_2) \left[\frac{1}{2} + \sin(\theta_2) \right] = \frac{1}{6} \left(10 \cdot \frac{\pi}{180} \right) \sin(\theta_2)$$

$$\therefore 1.0630 A_1 - 0.8529 A_3 = 0.02767$$

$$\therefore 1.7329 A_1 = 0.05066$$

$$\therefore A_1 = 0.02924$$

$$A_3 = 4.1321 \times 10^{-3}$$

$$\therefore \frac{C_L}{C_{Di}} = \frac{A_1}{A_1^2 + 3A_3^2} = 32.27$$

3. Calculate the particular elliptic circulation distribution $\Gamma(y)$ that gives an overall lift coefficient of $C_L = 0.8$ for an unswept, uncambered wing of aspect ratio $R = 8$, wing area S and semi-span s . Leave your answer in terms of the spanwise coordinate y , s , and U_∞ .

answer: $\Gamma(y) = \frac{2sU_\infty}{5\pi} \left(1 - \frac{y^2}{s^2}\right)^{\frac{1}{2}}$

$$\therefore C_L = \pi R A_1$$

$$\therefore A_1 = \frac{C_L}{\pi R} = \frac{1}{10\pi}$$

$$\therefore \Gamma_0 = 4sU_\infty A_1 = \frac{2sU_\infty}{5\pi}$$

$$\begin{aligned} \therefore \Gamma &= \Gamma_0 \sqrt{1 - \left(\frac{y}{s}\right)^2} \\ &= \frac{2sU_\infty}{5\pi} \sqrt{1 - (y/s)^2} \end{aligned}$$

for elliptical initial guesses,

$$\begin{aligned} L &= \rho U_\infty \int_{-s}^s \Gamma(y) dy = \rho U_\infty \int_{-1}^1 \Gamma_0 \sqrt{1 - \left(\frac{y}{s}\right)^2} d\frac{y}{s} \\ &= 2\rho U_\infty s \int_0^1 \Gamma_0 \sqrt{1 - \left(\frac{y}{s}\right)^2} d\frac{y}{s} \\ &= 2\rho U_\infty \Gamma_0 \left(\frac{\pi}{2}\right)s \\ &= \rho U_\infty \Gamma_0 \pi s \end{aligned}$$

$$\therefore L = \frac{1}{2} \rho U_\infty^2 S_{ref} C_L = \frac{1}{2} \rho U_\infty^2 (2s\bar{c} \times a_0 \alpha)$$

$$\therefore \Gamma_0 = \frac{\rho U_\infty^2 (\bar{c}) (a_0 \alpha)}{\rho U_\infty \pi s} = \frac{U_\infty \bar{c} a_0 \alpha}{\pi}$$

4. Calculate the overall lift coefficient for low speed flow over a plane untwisted wing of aspect ratio $R = 8$, elliptic planform, and NACA 652-415 section at 15° incidence to the undisturbed airstream, when operating at a chord Reynolds number of about 10^7 . Use the graph of section lift coefficient against incidence given in the attached figure and an iterative method of solution to obtain a reasonable estimate. Repeat the calculation for the wing at 24° incidence. Comment on the results.

answer: 1.39, 1.6

For elliptical, untwisted wing,

$$C_{Di} = \frac{C_L^2}{\pi R}, \quad W = -U_\infty \frac{C_L}{\pi R}$$

$$\therefore \alpha_i = -\frac{W}{U_\infty} = \frac{C_L}{\pi R}$$

$$\therefore \alpha = \alpha_e + \alpha_i = \alpha_e + \frac{C_L}{\pi R}$$

$$\therefore \alpha_e = \alpha - \alpha_i = \alpha - \frac{C_L}{\pi R} = \alpha - \frac{C_L}{8\pi}$$

$$\text{For } \alpha = 15^\circ = \frac{\pi}{12},$$

$$\alpha_e = \frac{\pi}{12} - \frac{C_L}{8\pi}$$

From graph-

$$\text{take } \alpha_e = 12^\circ = 0.209, \quad C_L = 1.4 \quad \therefore \frac{\pi}{12} - \frac{C_L}{8\pi} = 0.206 < 0.209$$

$$\therefore \text{take } \alpha_e = 11.5^\circ = 0.201, \quad C_L = 1.39 \quad \therefore \frac{\pi}{12} - \frac{C_L}{8\pi} = 0.206 > 0.201$$

.....

$$\rightarrow C_L = 1.39$$

$$\text{Similarly, when } \alpha = 24^\circ = \frac{2}{15}\pi, \quad C_L = 1.6$$

